Stochastic differential equations on Hilbert spaces

2.1. Mild, weak and strong solutions

Throughout the whole section we fix two separable (real) Hilbert spaces U, H and a Q-Wiener process $(W_t)_{t\geq 0}$ w.r.t. $(\mathcal{F}_t)_{t\geq 0}$ on U. Consider the equation

$$\begin{cases} dX_t &= [AX_t + B(X_t)]\, dt + C(X_t)\, dW_t \in H \\ X_0 &= \xi \end{cases} \tag{2.1}$$

with

(A.1) $(A, D(A))$ generates a C_0-semigroup $(T_t)_{t\geq 0}$ on H
(A.2) $B : H \to H$ is $\mathcal{B}(H)$-measurable
(A.3) $C : H \to L_2(U_0, H)$ is strongly continuous, i.e.,

$$x \mapsto C(x)u\,, H \to H$$

is continuous for all $u \in U_0$.
(A.4) ξ is H-valued and $\mathcal{F}_0$-measurable.

Notions of solutions

- **mild solution:** An H-valued predictable process $(X_t)_{t\in[0,T]}$ satisfying

$$X_t = T_t\xi + \int_0^t T_{t-s}B(X_s)\, ds + \int_0^t T_{t-s}C(X_s)\, dW_s \qquad \mathbb{P}-a.s.\,,$$

  for all $t \in [0, T]$, where all integrals have to be well-defined.
- **(analytically) strong solution:** A $D(A)$-valued predictable process $(X_t)_{t\in[0,T]}$ satisfying

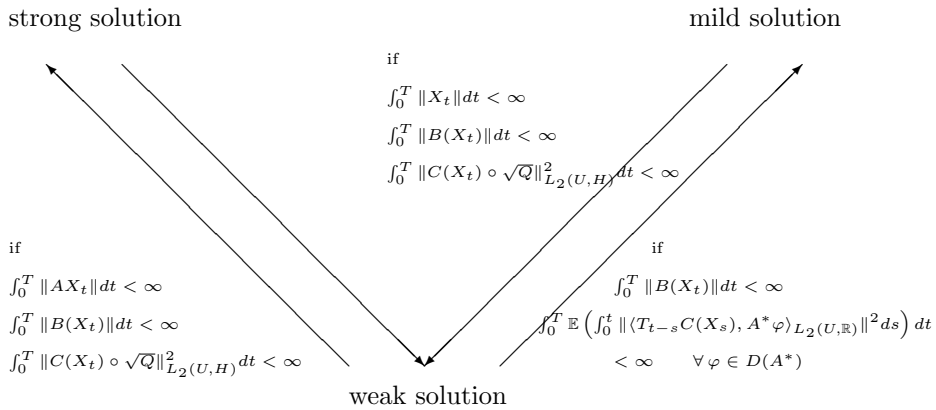
$$X_t = \xi + \int_0^t [AX_s + B(X_s)]\, ds + \int_0^t C(X_s)\, dW_s \qquad \mathbb{P}-a.s.\,,$$

  for all $t \in [0, T]$, where all integrals have to be well-defined.
- **(analytically) weak solution:** An H-valued predictable process $(X_t)_{t\in[0,T]}$ satisfying

$$\begin{aligned} \langle X_t, \varphi\rangle_H = \langle \xi, \varphi\rangle_H &+ \int_0^t \langle X_s, A^*\varphi\rangle_H + \langle B(X_s), \varphi\rangle_H\, ds \\ &+ \int_0^t \langle \varphi, C(X_s)\, dW_s\rangle_H \qquad \mathbb{P}-a.s.\,, \end{aligned}$$

  for all $t \in [0, T]$, $\varphi \in D(A^*)$. Here, $(A^*, D(A^*))$ is the dual operator of A and it is required that all integrals are well-defined.

<u>Interrelations between the different notions of solutions</u>



2.1.1. Existence and uniqueness of mild solutions. In this subsection we discuss existence and uniqueness of mild solutions of (2.1) under the following additional assumption:

$$\text{(H.1)} \qquad \|B(t,x) - B(t,y)\|_H + \|(C(t,x) - C(t,y)) \circ \sqrt{Q}\|_{L_2(U,H)} \le L\|x-y\|_H \quad \forall x,y \in H, t \in [0,T],$$

$$\text{(H.2)} \qquad \|B(t,x)\|_H^2 + \|C(t,x) \circ \sqrt{Q}\|^2_{L_2(U,H)} \le M\left(1 + \|x\|_H^2\right) \quad \forall x \in H, t \in [0,T].$$

THEOREM 2.1. *Under hypotheses (A.1)-(A.4), (H.1) and (H.2) there exists a unique mild solution of* (2.1) *satisfying*

$$\mathbb{P}\left(\int_0^T \|X_s\|_H^2 \, ds < \infty\right) = 1\,.$$

$(X_t)_{t\in[0,T]}$ *has a continuous modification and*

$$\mathbb{E}\left(\sup_{t\in[0,T]} \|X_t\|_H^p\right) \le c_{p,T}\left(1 + \mathbb{E}\left(\|\xi\|^p\right)\right) \qquad \forall p > 2\,.$$

PROOF. Application of Banach's fixed point theorem.

Fix $p \ge 2$ and define

$$\mathcal{H}_p := \left\{ Y : \Omega_T \longrightarrow H : Y \text{ predictable}, \|Y\|_p^p := \sup_{t\in[0,T]} \mathbb{E}\left(\|Y(t)\|_H^p\right)^{\frac{1}{p}} < \infty \right\}.$$

Then $(\mathcal{H}_p, \|\cdot\|_p)$ is a Banach space.

For $Y \in \mathcal{H}_p$ we define

$$\begin{aligned}\mathcal{K}(Y)(t) &:= T_t(\xi) + \int_0^t T_{t-s}(B(s,Y(s)))ds + \int_0^t T_{t-s}(C(s,Y(s)))dW(s)\\ &= T_t(\xi) + \mathcal{K}_1(Y)(t) + \mathcal{K}_2(Y)(t),\end{aligned}$$

with

$$\mathcal{K}_1(Y)(t) := \int_0^t T_{t-s}(B(s,Y(s)))ds$$

and

$$\mathcal{K}_2(Y)(t) := \int_0^t T_{t-s}(C(s,Y(s)))dW(s).$$

We will show that if $\mathbb{E}(\|\xi\|_H^p) < \infty$ this implies that $\mathcal{K}$ is a mapping

$$\mathcal{K} : \mathcal{H}_p \longrightarrow \mathcal{H}_p$$

and a strict contraction for T sufficiently small.

Concerning $\mathcal{K}_1$:

$$\begin{aligned}
\mathbb{E}(\|\mathcal{K}_1(Y)(t)\|_H^p) &= \mathbb{E}\left(\|\int_0^t T_{t-s}(B(s,Y(s)))ds\|_H^p\right) \\
&\overset{\text{Bochner}}{\leq} \mathbb{E}\left(\left(\int_0^t \|T_{t-s}(B(s,Y(s)))\|_H ds\right)^p\right) \\
&\overset{\text{Hölder}}{\leq} t^{p-1}\mathbb{E}\Bigg(\int_0^t \underbrace{\|T_{t-s}(B(s,Y(s)))\|_H^p}_{\leq M_1^p} \underbrace{\|B(s,Y(s))\|_H^p}_{\overset{(H.2)}{\leq} 2^p M^p(1+\|Y(s)\|_H^p)} ds\Bigg) \\
&\overset{t\leq T}{\leq} 2^p M_1^p M^p T^p (1+\|Y\|_p^p)
\end{aligned}$$

For the stochastic part $\mathcal{K}_2$ we will apply Burkholder-Davis-Gundy inequality:

$$\begin{aligned}
\mathbb{E}(\|\mathcal{K}_2(Y)(t)\|_H^p) &= \mathbb{E}\left(\|\int_0^t T_{t-s}(C(s,Y(s)))dW(s)\|_H^p\right) \\
&\leq c_p \mathbb{E}\left(\left(\int_0^t \|T_{t-s}(C(s,Y(s)))\circ\sqrt{Q}\|_{L_2(U,H)}^2 ds\right)^{\frac{p}{2}}\right) \\
&\leq c_p M_1^p \mathbb{E}\left(\left(\int_0^t \|C(s,Y(s))\circ\sqrt{Q}\|_{L_2(U,H)}^2 ds\right)^{\frac{p}{2}}\right) \\
&\underset{\substack{(H.2)\\ t\leq T}}{\overset{\text{Hölder}}{\leq}} c_p M_1^p 2^p M^p T^{\frac{p+1}{2}}(1+\|Y\|_p^p)
\end{aligned}$$

These estimates show that $\mathcal{K}(Y) \in \mathcal{H}_p$. To see that $\mathcal{K}$ is a strict contraction for small T, observe that similar to the above estimates

$$\begin{aligned}
\mathbb{E}(\|\mathcal{K}(Y_1)(t)-\mathcal{K}(Y_2)(t)\|_H^p) &\leq 2^p\left[\mathbb{E}\left(\|(\mathcal{K}_1(Y_1)-\mathcal{K}_1(Y_2))(t)\|_H^p\right) + \mathbb{E}\left(\|(\mathcal{K}_2(Y_1)-\mathcal{K}_2(Y_2))(t)\|_H^p\right)\right] \\
&\overset{(H.1)}{\leq} \underbrace{c_p(T^p + T^{\frac{p+1}{2}})}_{<1 \text{ for } T \text{ small}} \|Y_1 - Y_2\|_p^p.
\end{aligned}$$

The existence of a continuous modification will follow from Proposition 2.2 below.

□

2.1.2. Existence of a continuous modification. First assume that $\mathbb{E}(\|\xi\|_H^{2p}) < \infty$ for some $p > 1$, so that in particular $\mathbb{E}\left(\int_0^T \|X_t\|^{2p} dt\right) < \infty$. Let $\Phi(t) := C(t, X_t)$, then

$$\mathbb{E}\left(\int_0^T \|\Phi(t) \circ \sqrt{Q}\|_{L_2(U,H)}^{2p} dt\right) \leq M^{2p}\mathbb{E}\left(\int_0^T (1 + \|X_t\|^2)^p dt\right) < \infty.$$

The following proposition now implies that the process

$$\int_0^t T_{t-s} C(s, X_s) dW_s, \quad t \in [0, T],$$

has a continuous modification.

The general case $\mathbb{E}(\|\xi\|^{2p}) = \infty$ follows from approximating ξ with $\xi_n = \xi 1_{\{\|\xi\|_H \leq n\}}$.

PROPOSITION 2.2. *Let $p > 1$ and $\Phi \in \mathcal{L}^{2p}(\Omega_T, \mathcal{P}_T, \mathbb{P}_T; L_2)$, i.e.*

$$\mathbb{E}\left(\int_0^T \|\Phi(s) \circ \sqrt{Q}\|_{L_2(U,H)}^{2p} ds\right) < \infty.$$

Then there exists a constant C_T, such that

$$\mathbb{E}\left(\sup_{t\in[0,T]} \|\int_0^t T_{t-s}\Phi(s) dW_s\|_H^{2p}\right) \leq C_T \mathbb{E}\left(\int_0^T \|\Phi(s) \circ \sqrt{Q}\|_{L_2(U,H)}^{2p} ds\right). \tag{2.2}$$

Moreover, $W_{A,\Phi}(t) := \int_0^t T_{t-s}\Phi(s) dW_s$ has a continuous modification.

PROOF. We will use a technique, called the **factorization method**, based on the following integral identity:

$$\int_s^t (t-r)^{\alpha-1}(r-s)^{-\alpha} ds = \frac{\pi}{\sin(\pi\alpha)} \quad \forall \alpha \in \left(0, \frac{1}{2}\right) \forall s < t.$$

Indeed, substituting $u = \frac{r-s}{t-s}$ yields

$$\int_s^t (t-r)^{\alpha-1}(r-s)^{-\alpha} dr = \int_0^1 (1-u)^{\alpha-1} u^{-\alpha} du = \frac{\Gamma(\alpha)\Gamma(1-\alpha)}{\Gamma(1)} = \frac{\pi}{\sin(\pi\alpha)} =: \frac{1}{c_\alpha}.$$

We can thus rewrite the convolution

$$
\begin{aligned}
W_{A,\Phi}(t) &= \int_0^t T_{t-s}(\Phi(s))dW(s) \\
&= c_\alpha \int_0^t \int_s^t (t-r)^{\alpha-1}(r-s)^{-\alpha} dr T_{t-s}(\Phi(s))dW(s) \\
&= c_\alpha \int_0^t \int_0^r (t-r)^{\alpha-1}(r-s)^{-\alpha} \underbrace{T_{t-s}(\Phi(s))}_{(T_{t-r}\circ T_{r-s})(\Phi(s))} dW(s)dr \\
&= c_\alpha \int_0^t (t-r)^{\alpha-1} T_{t-r} \underbrace{\left(\int_0^r (r-s)^{-\alpha} T_{r-s}(\Phi(s))dW(s) \right)}_{=:y(r)} dr.
\end{aligned}
\tag{2.3}
$$

Here we used in the third equality that we can interchange the order of integration w.r.t. r and s ("stochastic Fubini theorem"), and in the fourth step we have used linearity of (T_t).

Hence for $p > \frac{1}{2\alpha} > 1$ it follows that

$$
\begin{aligned}
\|W_{A,\Phi}(t)\|_H^{2p} &= c_\alpha^{2p} \| \int_0^t (t-r)^{\alpha-1} T_{t-r}(y(r))dr \|_H^{2p} \\
&\leq c_\alpha^{2p} \left(\int_0^t (t-r)^{\alpha-1} \|T_{t-r}(y(r))\|_H dr \right)^{2p} \\
&\leq c_\alpha^{2p} \left(\int_0^t (t-r)^{\frac{2p}{2p-1}(\alpha-1)} dr \right)^{2p-1} \int_0^t \|T_{t-r}(y(r))\|_H^{2p} dr \\
&\leq c_{\alpha,p,T} \int_0^t \|T_{t-r}(y(r))\|_H^{2p} dr
\end{aligned}
$$

Here we used Bochner's inequality in the first step, then Hölder's inequality in the second step (applied to $2p$ and $\frac{2p}{2p-1}$). In the last step we used the fact that

$$
p > \frac{1}{2\alpha} \Leftrightarrow \frac{2p}{2p-1}(\alpha-1) > \frac{2p}{2p-1}\left(\frac{1}{2p}-1\right) = -1.
$$

It follows that

$$
\sup_{t\in[0,T]} \|W_{A,\Phi}(t)\|_H^{2p} \leq c_{\alpha,p,T} \int_0^T \|y(r)\|_H^{2p} dr. \tag{2.4}
$$

The Burkholder-Davis-Gundy inequality implies that

$$
\mathbb{E}\left(\|y(r)\|_H^{2p} \right) \leq C_p \mathbb{E}\left(\left(\int_0^r (r-s)^{-2\alpha} \|\Phi(s)\circ\sqrt{Q}\|_{L_2}^2 ds \right)^p \right),
$$

hence

$$\begin{aligned}\int_0^T \mathbb{E}\left(\|y(r)\|_H^{2p}\right) dr &\le C_p \mathbb{E}\left(\int_0^T \left(\int_0^r (r-s)^{-2\alpha} \|\Phi(s)\circ\sqrt{Q}\|_{L_2}^2\, ds\right)^p dr\right)\\ &\le C_{\alpha,p,T}\mathbb{E}\left(\int_0^T\int_0^r (r-s)^{-2\alpha}\|\Phi(s)\circ\sqrt{Q}\|_{L_2}^{2p}\, ds\, dr\right)\\ &\le \tilde{C}_{\alpha,p,T}\mathbb{E}\left(\int_0^T \|\Phi(s)\circ\sqrt{Q}\|_{L_2}^{2p}\, ds\right).\end{aligned} \tag{2.5}$$

In the second step we have used Hölder's inequality, applied to the measure $(r-s)^{-2\alpha}\, ds$ on $[0,r]$, since

$$\begin{aligned}\left(\int_0^r (r-s)^{-2\alpha}\|\Phi(s)\circ\sqrt{Q}\|_{L_2}^2\, ds\right)^p \le \Big(\underbrace{\int_0^r (r-s)^{-2\alpha}\, dr}_{=\frac{r^{1-2\alpha}}{1-2\alpha}\le\frac{T^{1-2\alpha}}{1-2\alpha}}\Big)^{p-1}\cdot\\ \cdot\int_0^r (r-s)^{-2\alpha}\|\Phi(s)\circ\sqrt{Q}\|_{L_2}^{2p}\, ds\,.\end{aligned}$$

Combining with (2.4) implies

$$\mathbb{E}\left(\sup_{t\in[0,T]}\|W_{A,\Phi}(t)\|_H^{2p}\right) \le C_{\alpha,p,T}\mathbb{E}\left(\int_0^T \|\Phi(s)\circ\sqrt{Q}\|_{L_2}^{2p}\, ds\right).$$

We will show next that the mapping

$$F : y \mapsto \int_0^t (t-r)^{\alpha-1} T_{t-r} y(r)\, dr$$

maps $L^{2p}([0,T];H)$ into $C([0,T];H)$ which will imply together with (2.3) and (2.5) that $W_{A,\Phi}$ has a continuous modification.

Note that similar to the proof of (2.4)

$$\sup_{t\in[0,T]}\|F(y)(t)\|_H^{2p} \le c_{\alpha,T,p}\int_0^T \|y(t)\|_H^{2p} dt.$$

Since $C([0,T];H) \subset L^{2p}([0,T];H)$ is dense, it therefore suffices to show that $t\mapsto F(y)(t)$ is continuous, provided $t\mapsto y(t)$ is continuous. Indeed, we can approximate $y\in L^{2p}([0,T];H)$ by a sequence $(y_n)\subset C([0,T];H)$ converging to y in $L^{2p}([0,T];H)$. Then $F(y_n)$ converges uniformly to $F(y)$, which implies that $F(y)$ is continuous too.

So assume that $t \mapsto y(t)$ is continuous. We will then show that $F(y)$ is right-/left-continuous:

$$
\begin{aligned}
&\|F(y(t+h)) - F(y(t))\|_H \\
&\quad = \left\| \int_0^{t+h} (t+h-r)^{\alpha-1} T_{t+h-r}(y(r))dr - \int_0^t (t-r)^{\alpha-1} T_{t-r}(y(r))dr \right\|_H \\
&\quad \leq \left\| \int_0^t (t+h-r)^{\alpha-1} T_{t-r}(T_h(y(r)))dr - \int_0^t (t-r)^{\alpha-1} T_{t-r}(y(r))dr \right\|_H \\
&\qquad\qquad + \left\| \int_t^{t+h} (t+h-r)^{\alpha-1} T_{t+h-r}(y(r))dr \right\|_H \\
&\quad \leq \int_0^t \left\| \left((t+h-r)^{\alpha-1} T_h - (t-r)^{\alpha-1} \right) T_{t-r}(y(r)) \right\|_H dr \\
&\qquad\qquad + \int_t^{t+h} (t+h-r)^{\alpha-1} \|T_{t+h-r}(y(r))\|_H dr \\
&\quad \xrightarrow{h \searrow 0} 0,
\end{aligned}
$$

since $\left((t+h-r)^{\alpha-1} T_h - (t-r)^{\alpha-1} \right) T_{t-r}(y(r)) \longrightarrow 0$ in H and bounded in the H-norm by $2(t-r)^{\alpha-1}$ up to some uniform constant, and $(t+h-r)^{\alpha-1} T_{t+h-r} y(r)$ also bounded in the H-norm up to some uniform constant by $(t+h-r)^{\alpha-1}$.

The left-continuity is shown in a similar way, hence the assertion is proven. $\square$